

Learned Generators for Structure-Compatible Generalization

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Phase 3: finite generator discovery plus transfer

Abstract

Earlier structure-compatible generalization results used oracle deployment transformations or finite shifts inferred from observed overlaps. Phase three learns finite generator families from data: modular affine input/label transports and vision rotations selected by training-set self-consistency. The result is a bounded test of OOD prediction and intervention without OOD labels, not an open-world transformation-discovery theorem.

1. Regime Transition

The old regime supplied the transformation parameterization. The new operation learns candidate transports from data and evaluates whether the learned family predicts or controls OOD behavior under ID-equivalent training evidence.

Field	Value
GPU	L4
modular configs	90
vision base units	36
modular epochs	350
vision epochs	220
rows	416

Table 1. Phase-three Modal L4 manifest.

2. Predictor Correlations

Domain	Predictor	Pearson r	Spearman r	N
modular_learned_generator	compatibility_true	0.812	0.367	270
modular_learned_generator	compatibility_discovered	0.787	0.474	270
modular_learned_generator	negative_train_loss	0.432	0.250	270
modular_learned_generator	id_validation_accuracy	0.398	0.120	270
modular_learned_generator	train_accuracy	0.398	0.120	270
modular_learned_generator	negative_abs_sharpness	0.050	0.224	270
modular_learned_generator	compatibility_wrong	-0.085	0.049	270
modular_learned_generator	negative_parameter_l2	-0.448	-0.230	270
vision_rotation_learned_generator	compatibility_true	0.678	0.591	144
vision_rotation_learned_generator	id_validation_accuracy	0.661	0.632	144
vision_rotation_learned_generator	train_accuracy	0.661	0.632	144
vision_rotation_learned_generator	negative_train_loss	0.659	0.514	144
vision_rotation_learned_generator	compatibility_discovered	0.599	0.502	144
vision_rotation_learned_generator	compatibility_wrong	0.457	0.341	144

Table 2. Phase-three predictors against OOD.

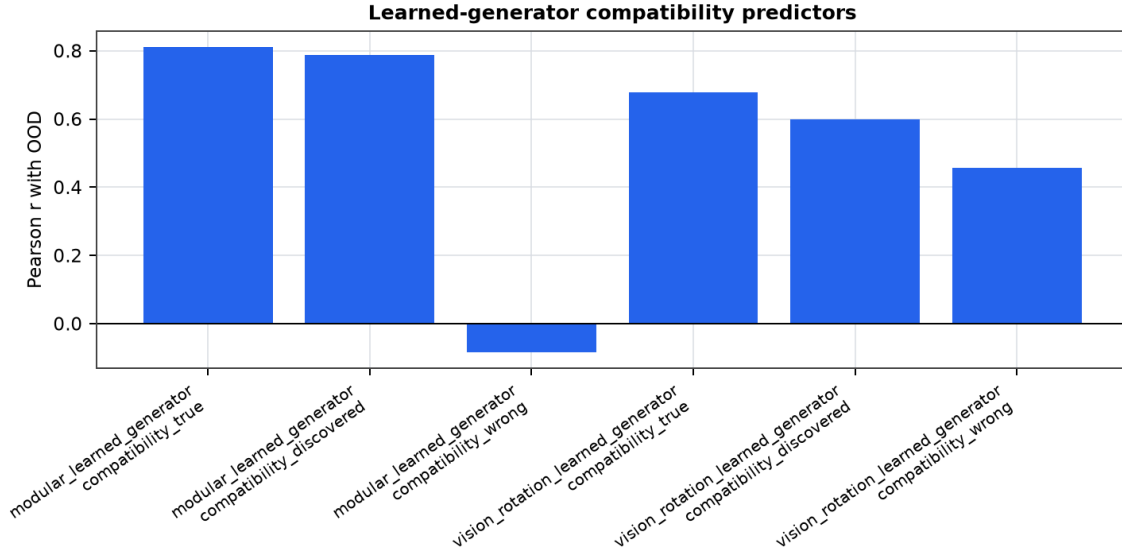


Figure 1. Learned-generator compatibility predictors.

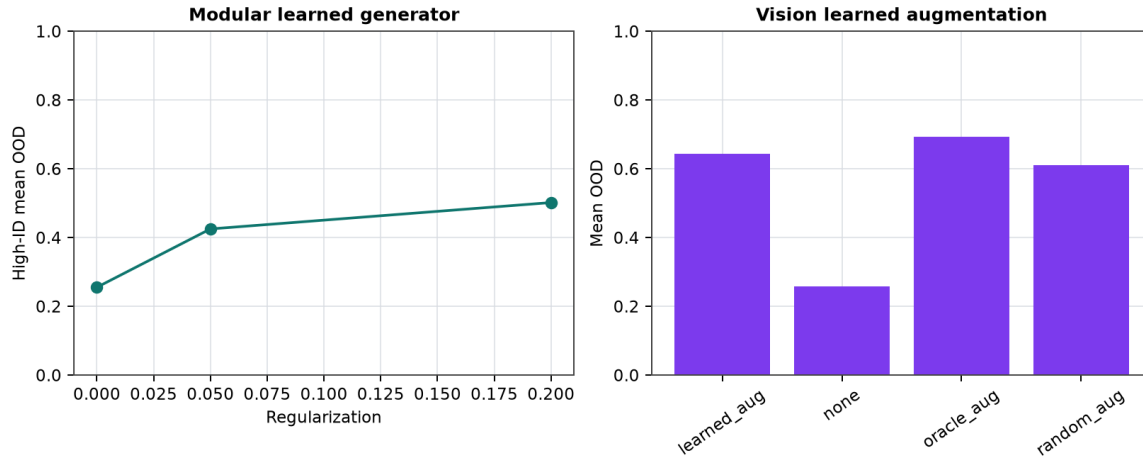


Figure 2. Learned-generator interventions.

3. Interventions

Reg	N	Mean train	Mean OOD	High-ID N	High-ID OOD
0.000	90	0.652	0.168	45	0.256
0.050	90	0.653	0.253	45	0.425
0.200	90	0.656	0.290	45	0.502

Table 3. Modular learned-generator regularization arms.

Regime	N	Mean train	Mean OOD	Learned compat
learned_aug	36	0.707	0.644	0.793
none	36	0.815	0.258	0.486
oracle_aug	36	0.709	0.693	0.801

random_aug	36	0.679	0.610	0.760
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Table 4. Vision rotation transfer arms.

4. Scope

The result is a learned finite-generator transfer result. It is stronger than hand-given oracle groups, but the generator classes are still chosen by the experimenter. Language/template substitution is the next unclaimed transfer step.